How to Fine-Tune LLMs for Domain-Specific Adaptation

This second article in the two-part series presents a step-by-step approach to fine-tuning a large language model for domain-specific knowledge.

As discussed in the previous article carried in the March 2026 issue of *OSFY,* large language models (LLMs) possess broad general knowledge but often lack the depth and reliability required for specialised or proprietary domains.

In this second article, we discuss and demonstrate in detail how to adapt a general-purpose base LLM into a domain-specialised coding assistant for C and Linux system programming tasks using fine-tuning techniques such as LoRA and QLoRA. The approach illustrated can be extended to fine-tune and adapt a base LLM for proprietary hardware architectures, enabling its effective use as a coding assistant for system software development tasks.

Practical blueprint for fine-tuning Mistral-7B for coding assistance in C and Linux system programming

This blueprint demonstrates the end-to-end fine-tuning of the Mistral-7B model on a free-tier Google Colab T4 GPU. To fit within the T4's 16GB VRAM, the model is loaded in 4-bit form using the bitsandbytes library, reducing memory usage while preserving performance. This enables efficient forward and backward passes, and makes parameter-efficient methods such as LoRA and QLoRA practical on constrained hardware. Only the lightweight adapter parameters are updated during training, ensuring stable fine-tuning within strict GPU memory limits. We have broadly classified the entire fine-

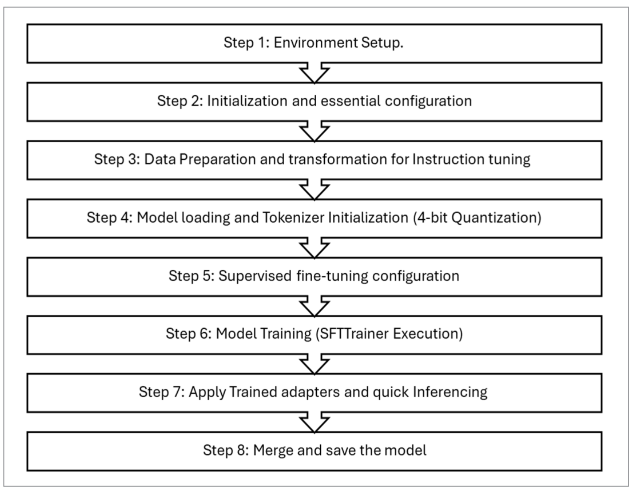

Figure 1: End-to-end supervised fine-tuning